Detecting Fake Reviewer Communities in Amazon Product Reviews



Research question

Is it possible to detect suspicious rating behaviors and coordinated review coalitions within the Amazon ecosystem?



Data Structure & EDA

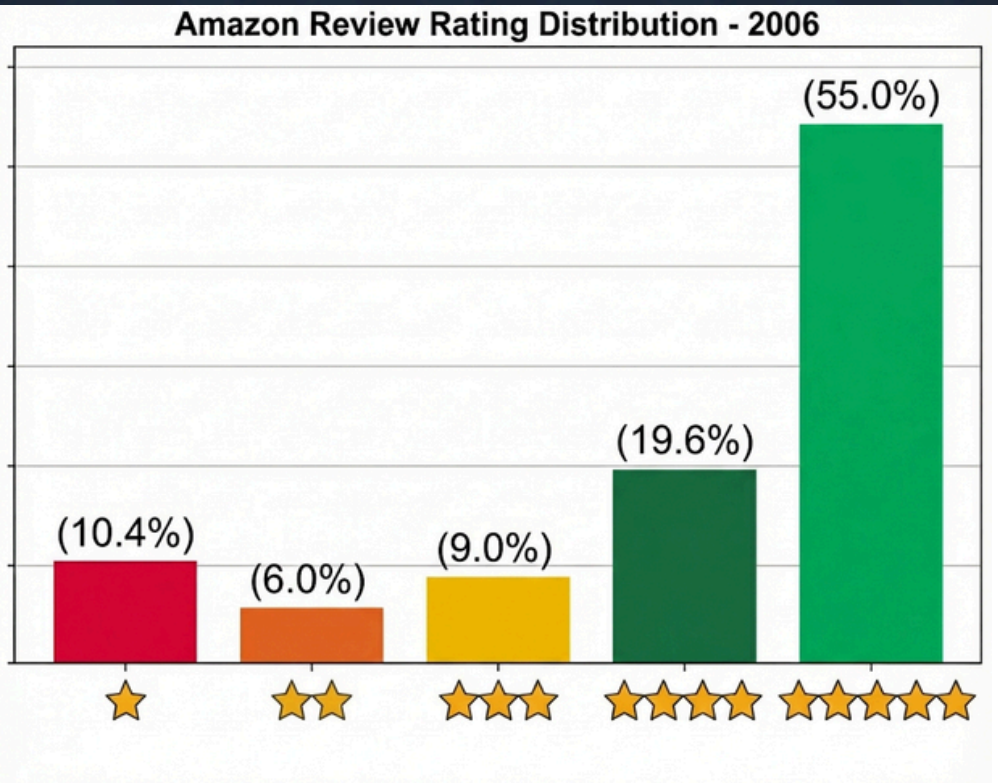
- **Data Structure:** Original format with four columns: User ID | Product ID | Rating | Timestamp (Unix Epoch, then converted to simple format dd/mm/yyyy)
- The dataset contains 2.9 million reviews distributed between the years 1996-2006

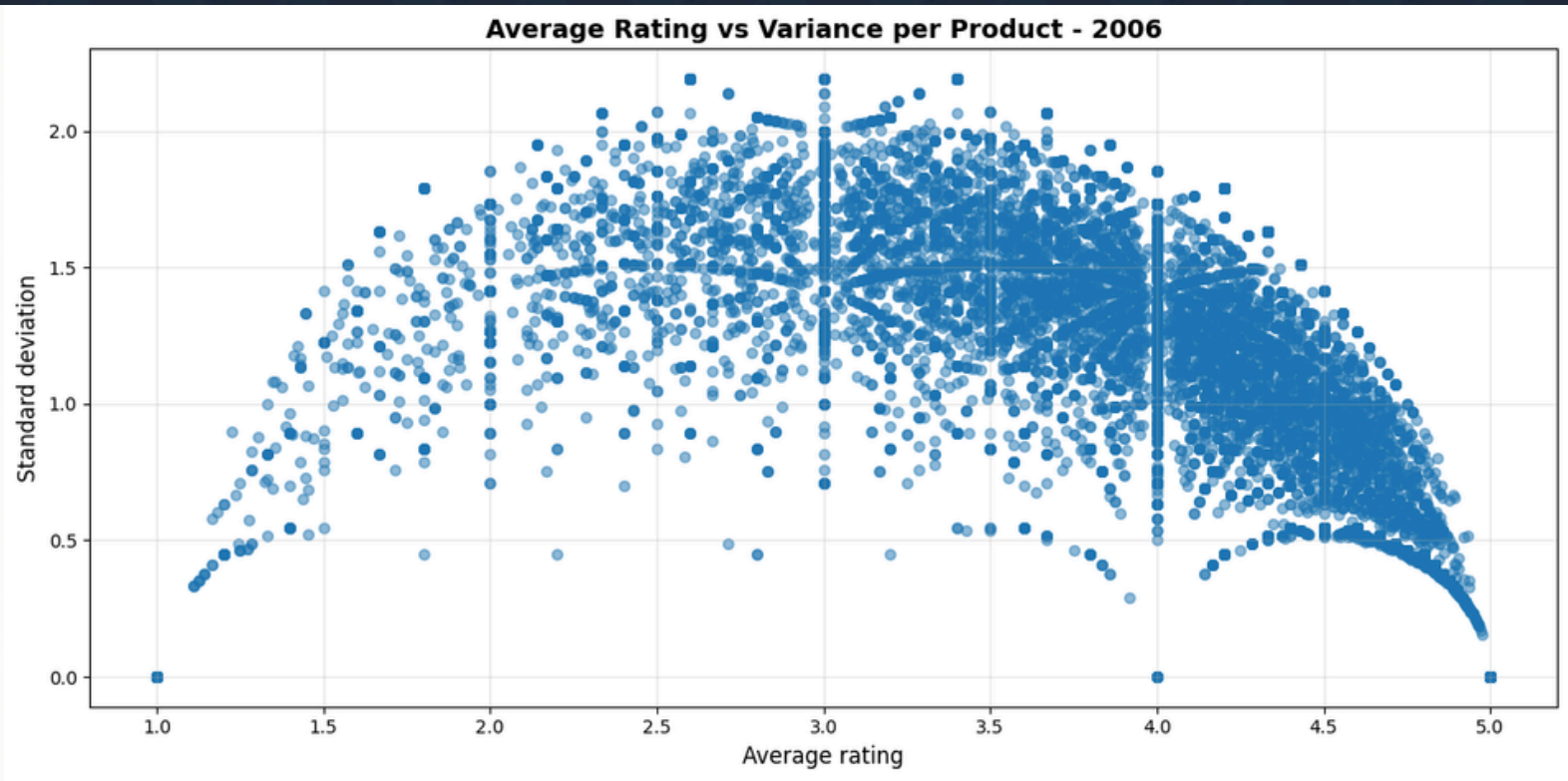
1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006
11	8852	46617	127887	339762	308792	311811	340743	422446	581195	260349

- In this analysis we focused on data belonging to the year 2006
- **Source:** Amazon.com (via J. McAuley, J. Leskovec) from The Network Data Repository

Data Structure & EDA

- We performed an Exploratory Data Analysis on the year 2006: The dataset includes 517,684 reviews from 260,349 unique users and 230,332 unique products, with an average rating of 4.03 stars.
- A clear trend emerges in the rating distribution: There is a **strong positive bias**, with 5-star ratings dominating the platform. This indicates overall user satisfaction, though it also suggests potential rating inflation.

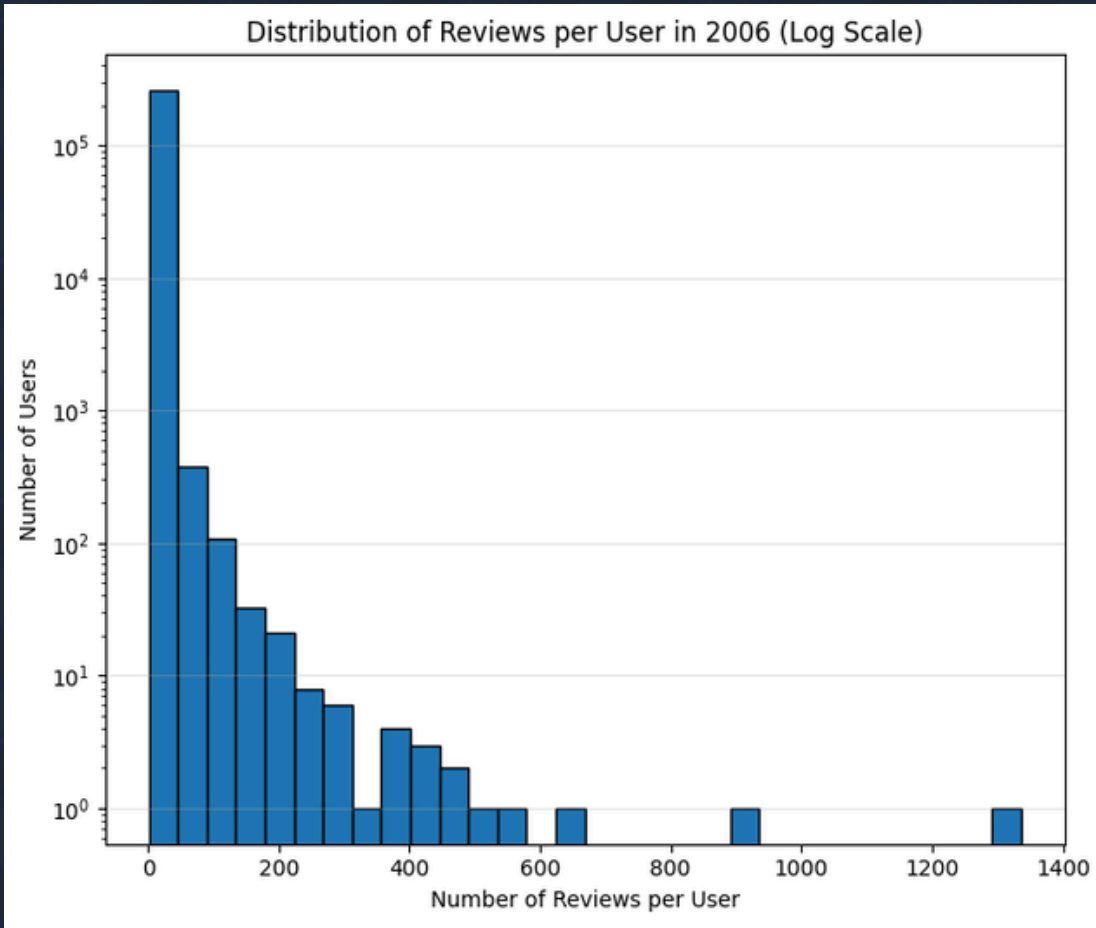


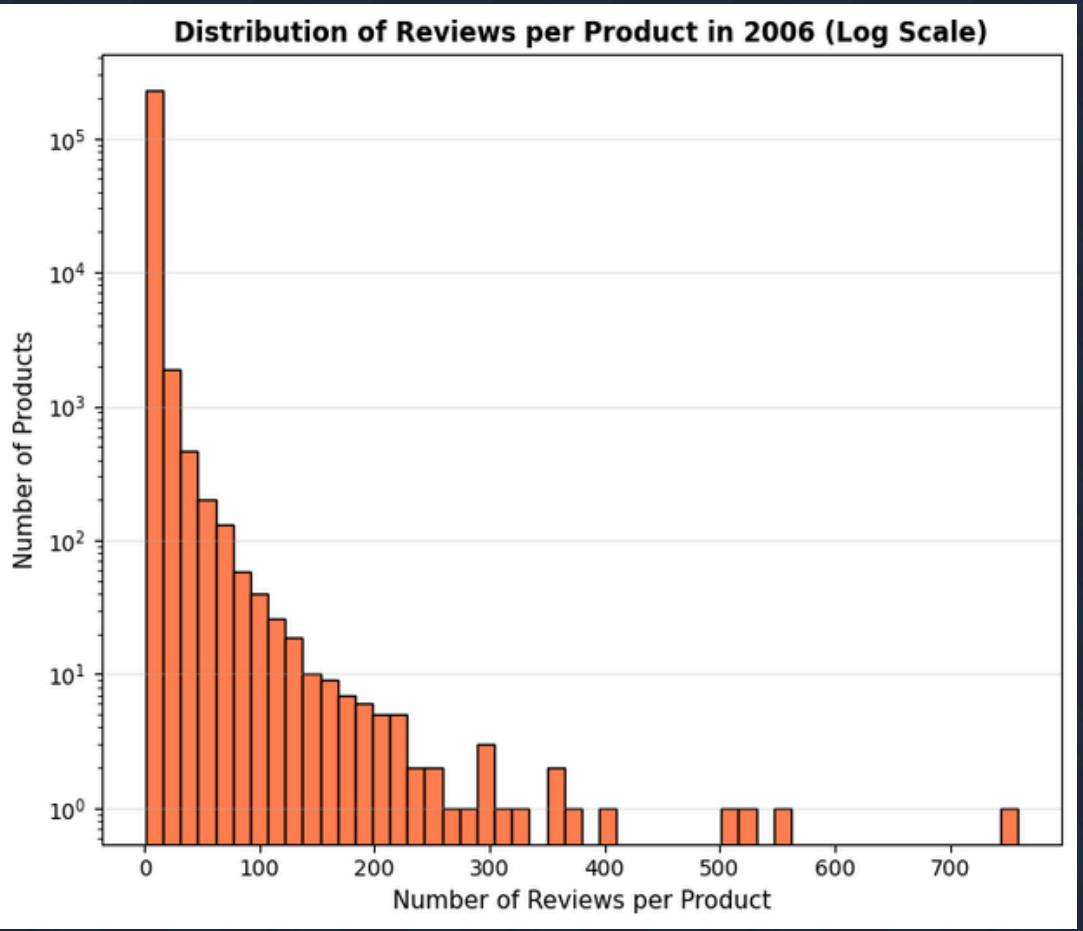


Data Structure & EDA

The data is massive but sparse: 73.7% of users wrote only one review and users with >10 reviews are just 1.7% .

69.5% of products received only one review and only 0.9% received more than 20 reviews.





Preprocessing & Network Refinement



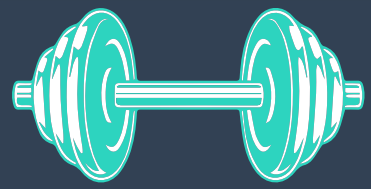
Graph creation

Created bipartite graph (users-products) from the dataset



Projection

Transformed bipartite graph into user-user projection



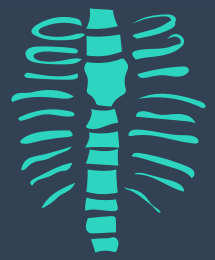
Edge Attributes

Edge weights based on number of co-reviewed products and rating similarity (Pearson)



Simplify

Removed isolated nodes and cycles



Backbone

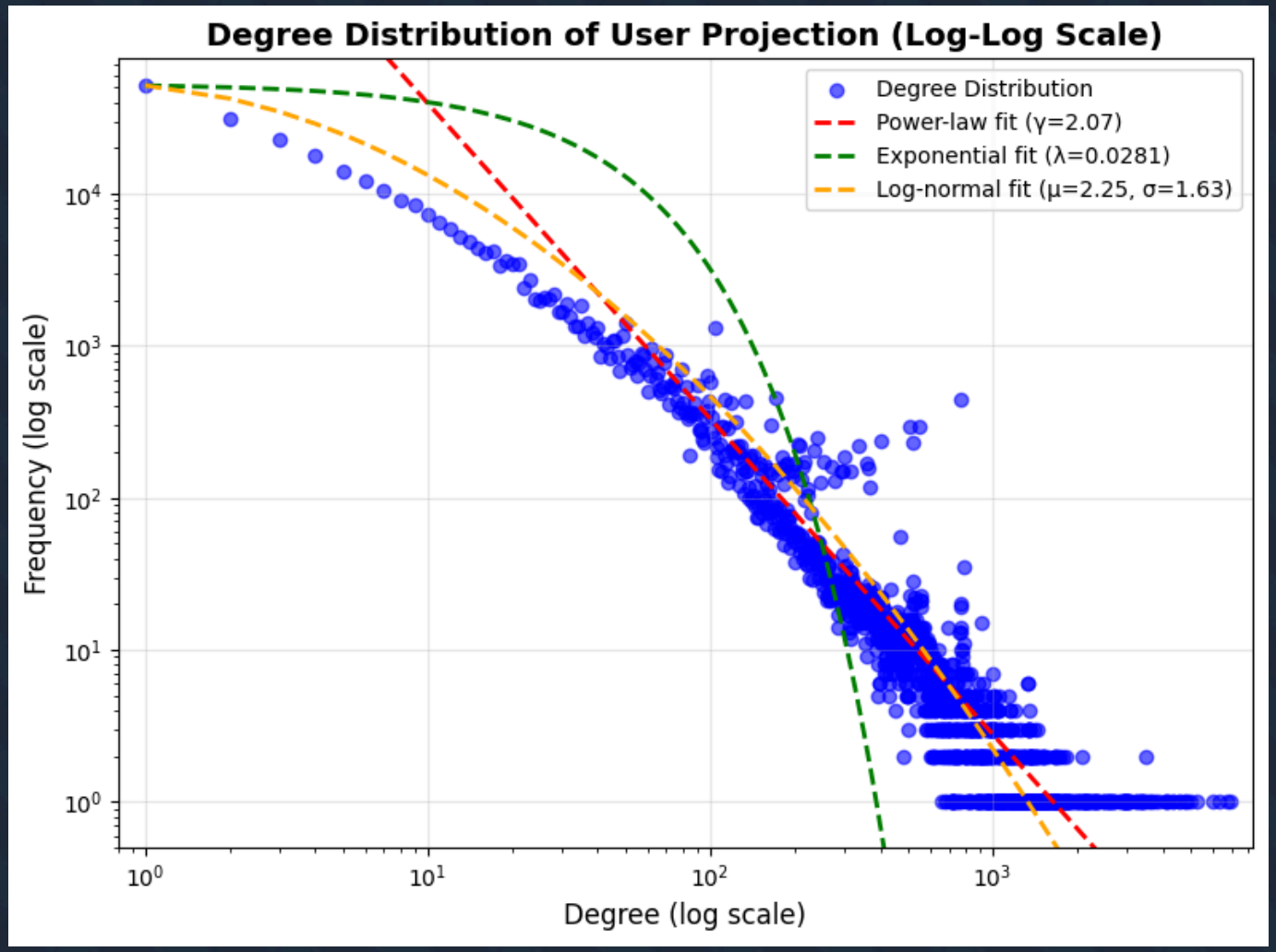
Applied backbone extraction to retain only statistically significant edges

- Nodes: 375k
- Edges: 6.69m
- Average Degree: 35.6
- Log of # nodes: 12.8

Network Analysis and Model Fitting

- The user projection network exhibits a power-law degree distribution with an estimated exponent $\gamma \approx -2.07$, characteristic of **scale-free networks**.

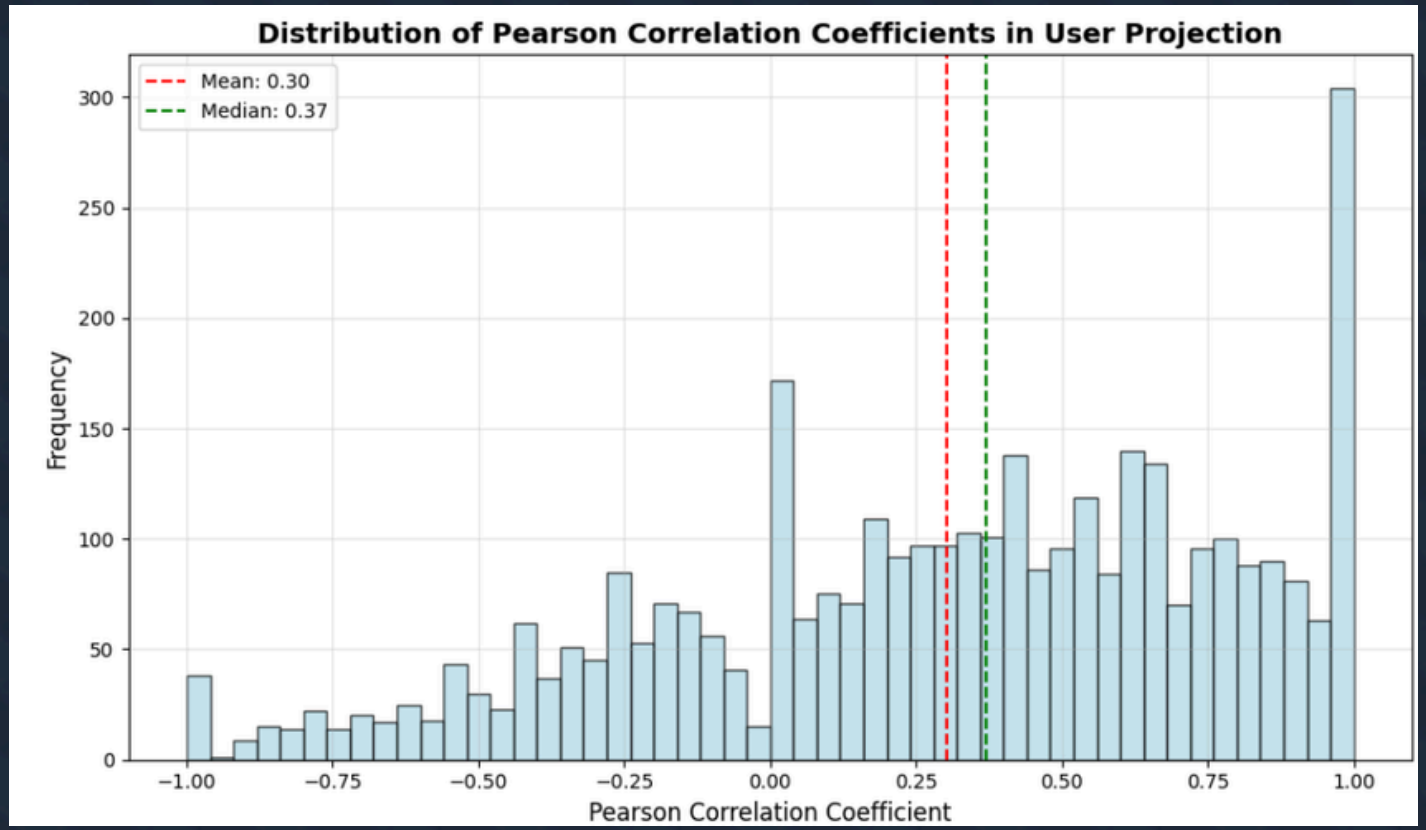
- This indicates that **most users have few connections** (co-reviewed products with other users), while a small number of ***hub* users** have disproportionately many connections.



User Similarity & Coordination Metrics

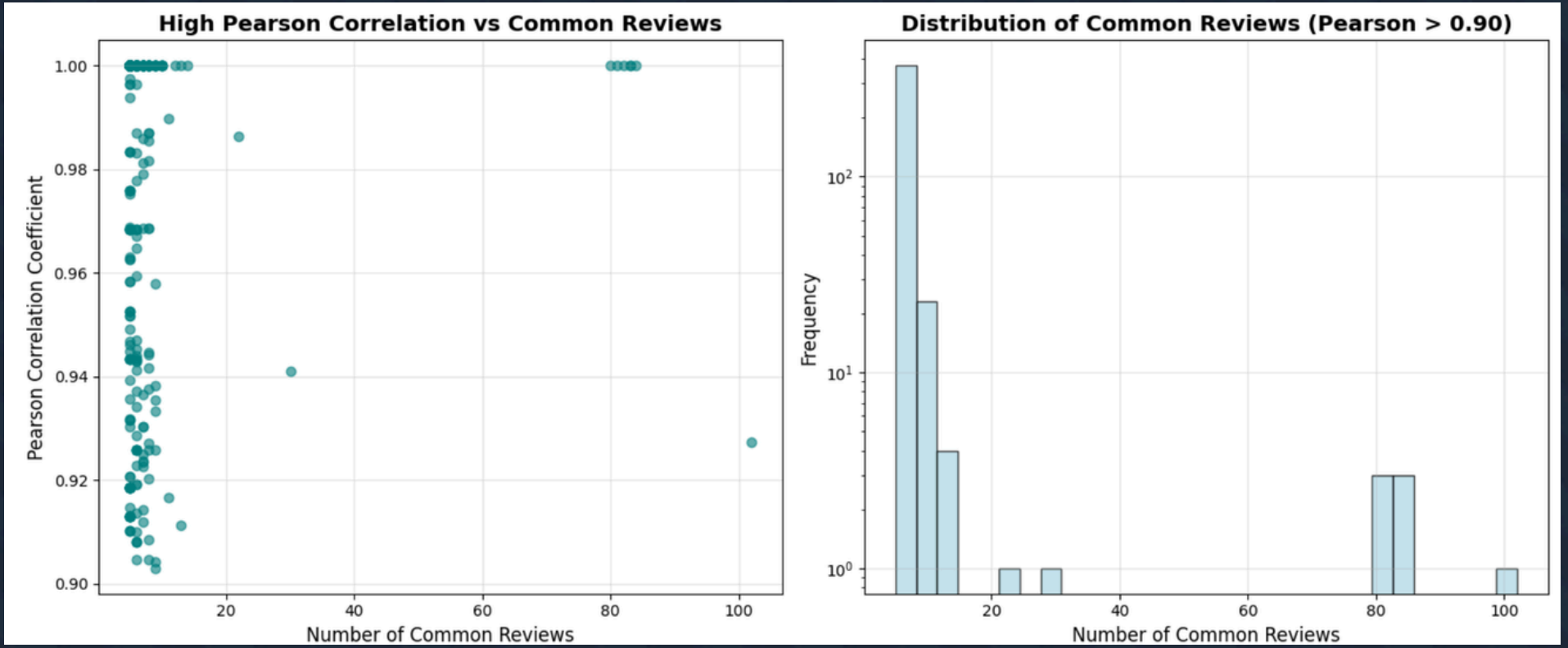
- Computed the **Pearson Correlation Coefficient (ρ)** on edges where nodes (users) shared 5 or more reviewed products.

- This method is particularly effective to measure **similarity between users**, accounting for similar rates that could potentially evolve into **coordinated rates**.



User Similarity & Coordination Metrics

It turned out that there are 406 edges with Pearson correlation>0.99, but the distribution of common reviews is **highly asymmetric** (this is pretty understandable, but we will come back to this result later)



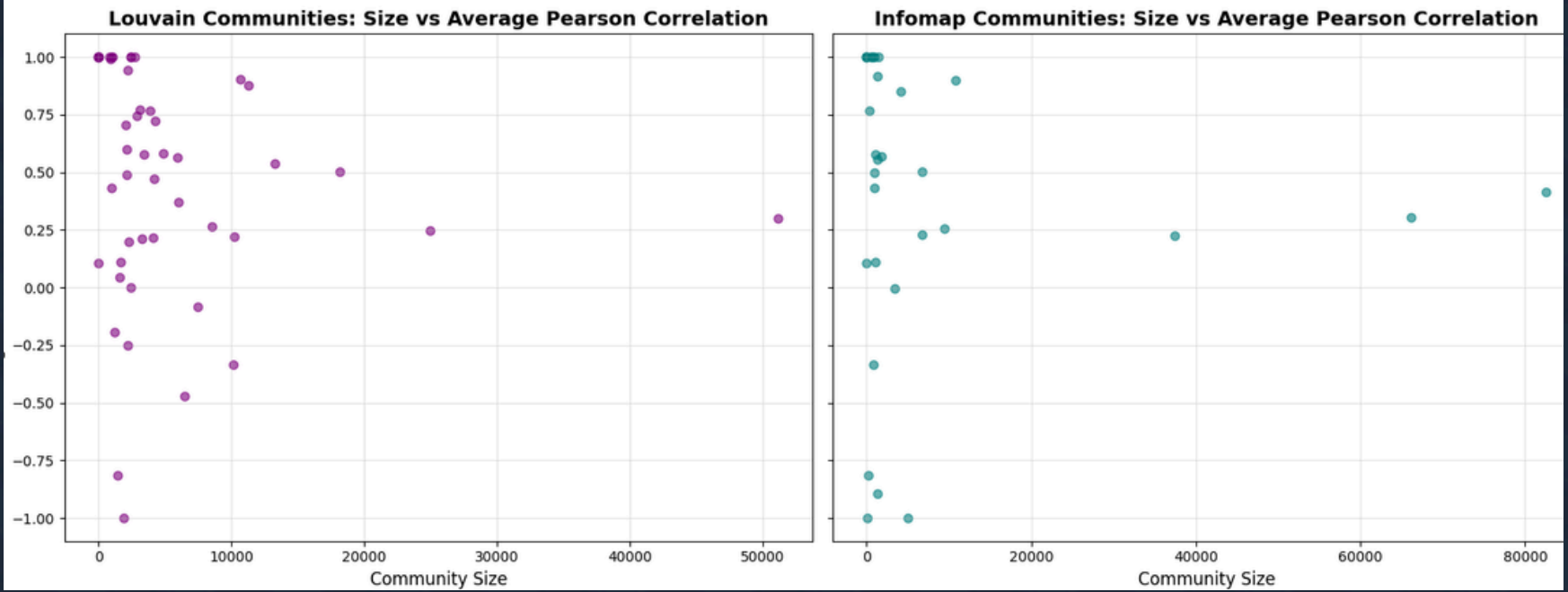
Community Detection & Algorithmic Comparison

To identify hidden coalitions, we applied three heuristic community detection algorithms on the User Projection graph. The edges were weighted based on the number of shared products between users.

1. **Louvain Algorithm:** A greedy optimization algorithm that seeks to maximize Modularity. It is efficient for large networks but can sometimes suffer from a resolution limit, failing to identify small communities.
2. **Leiden Algorithm:** A more advanced version of Louvain designed to fix its shortcomings. It guarantees that communities are well-connected and provides higher quality, more stable partitions.
3. **Infomap Algorithm:** A method based on the Map Equation and information theory. It uses random walks to analyze the information flow across the network, identifying clusters where flow remains trapped (ideal for detecting tightly coordinated groups).

Method	Number of Community	Modularity	Internal Density
Louvain	51970	0.75	0.18
Leiden	52068	0.02	0.01
InfoMap	10001	0.65	0.95

Refining Suspicious Coalitions



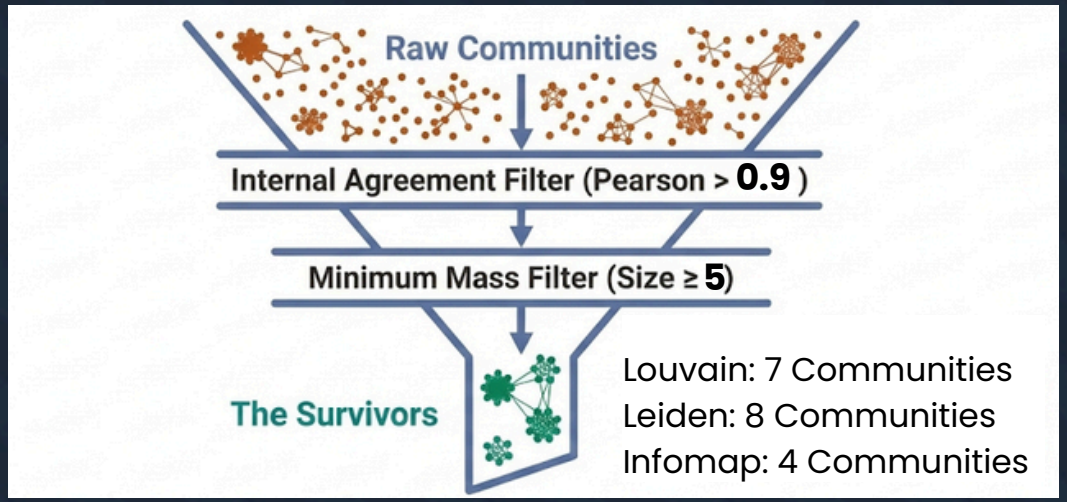
The objective is to filter the noise to isolate highly synchronized groups by cross-referencing Community Size and Internal Pearson Correlation.

Filtering Thresholds:

Mean Internal Correlation > 0.9: high rating synchronization.

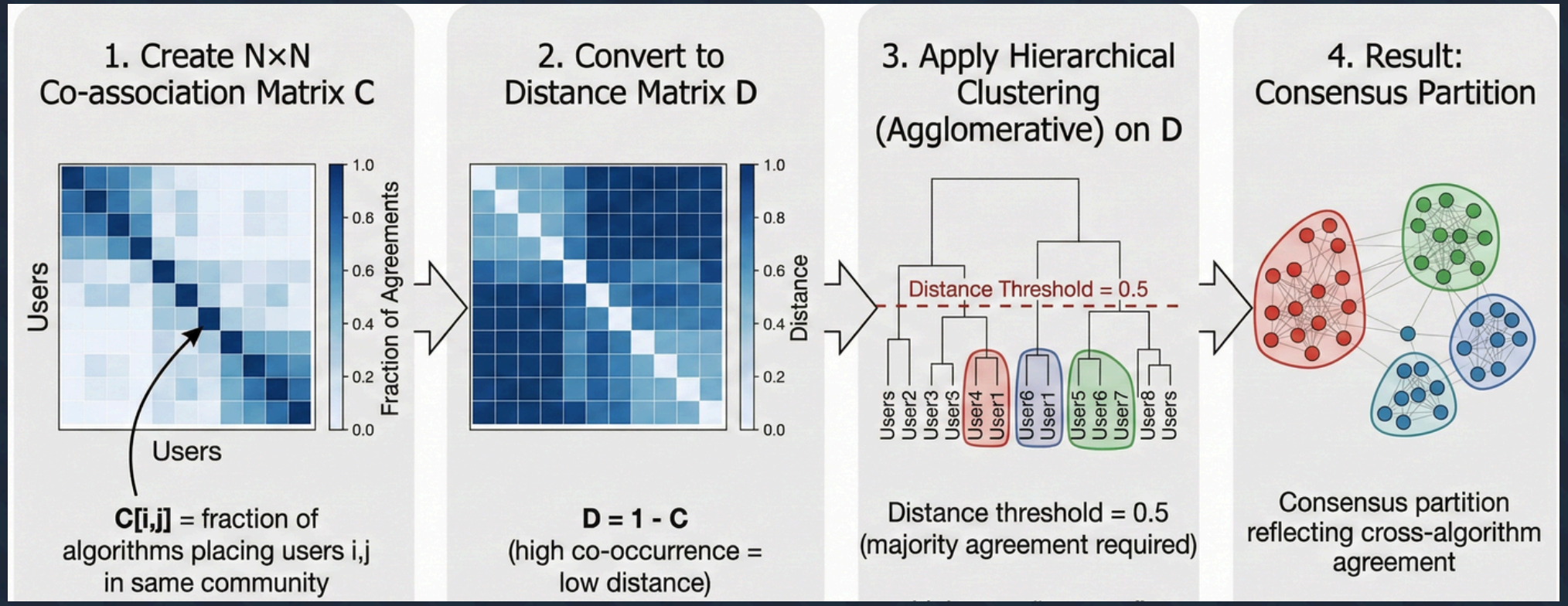
Community Size > 5: evidence of organized group activity.

We successfully filtered out noise and narrowed the search space to a few highly synchronized clusters.



Refining Detection: Dual-Consensus Strategy

Given the poor performance of the Leiden algorithm (few modularity), it was excluded from the consensus phase to avoid introducing noise. A dual-consensus approach was applied combining Louvain (structural density) and Infomap (information flow) to identify only the most robust community cores validated by both topological and dynamic metrics.

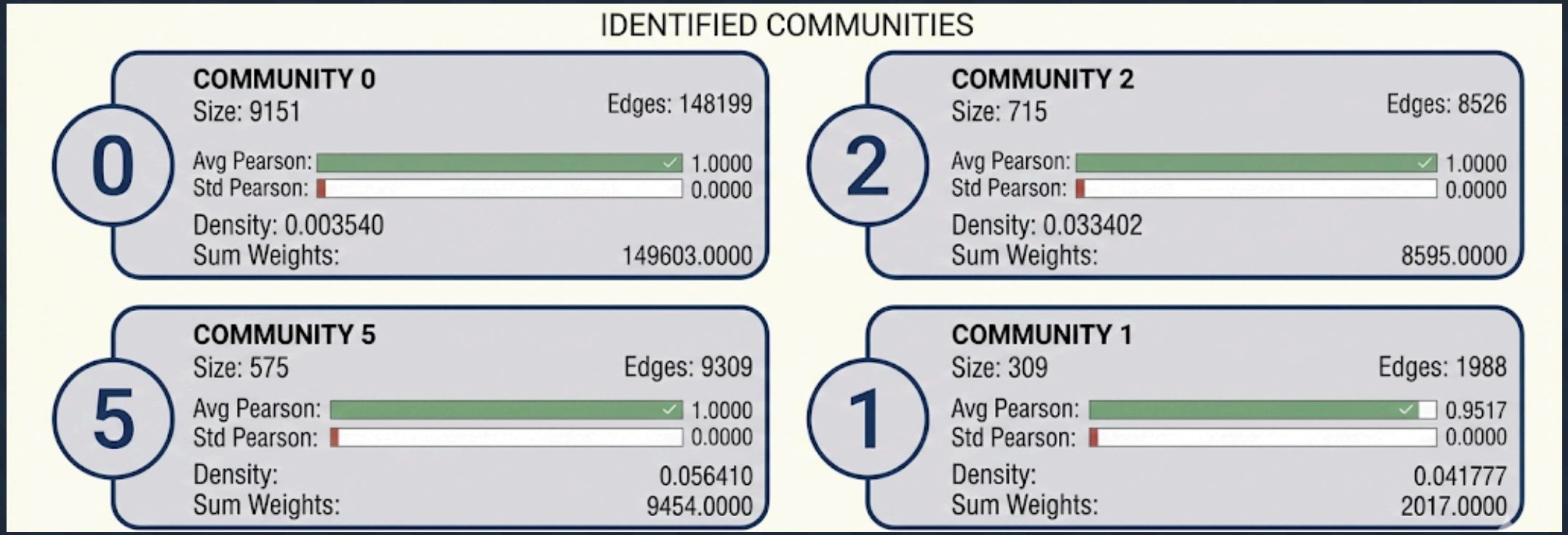


Consensus Results & Final Validation

Suspicious Community Ranking

Following the dual-consensus fusion, we identified the final clusters that strictly satisfy our suspicious criteria.

These communities represent the most coordinated user groups in the network.



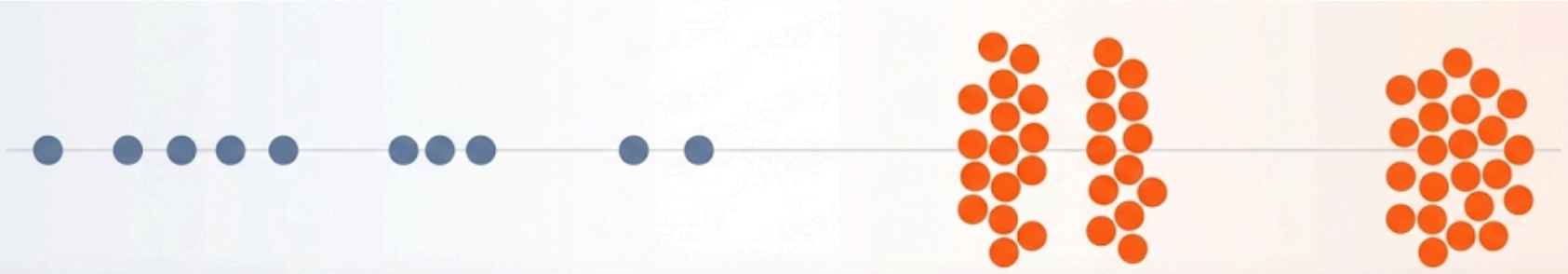
Temporal analysis: Phase 1 (Burstiness Detection)

For each product reviewed by suspicious communities members we calculated **τ** (tau), the time interval in hours between consecutive reviews. This value is then used to calculate both global and local **Burstiness Coefficient (B)**

$$B = \frac{\sigma_\tau - \mu_\tau}{\sigma_\tau + \mu_\tau}$$

σ_τ = Standard Deviation of time between reviews

μ_τ = Mean time between reviews



B → 0: Natural behavior

B → 1: Suspicious coordination

POISSON / RANDOM
Distributed irregularly over time

BURSTY / ATTACK
Clustered in short, dense intervals

Temporal analysis: Phase 1 (Burstiness Detection)

These are the top community-rated products by local Burstiness Coefficient (>0.1)

COMPLETE SUSPICIOUS PRODUCTS TABLE

Summary Stats: 146 Total Rows | 146 Unique Products | 4 Communities Involved

product_id	community_id	burstiness_local	reviews_local	mean_rating_local
842072	5	0.3784	8	5.0000
266549	0	0.3686	5	4.3750
842070	5	0.3591	5	5.0000
10710	0	0.3416	102	4.0367
842074	5	0.3229	5	5.0000
834149	5	0.3158	4	5.0000
301452	0	0.3084	60	3.0952
738921	0	0.3142	7	5.0000
842077	5	0.3023	15	5.0000
130429	0	0.2750	4	2.5000
...	...	...	...	...
516550	0	0.1160	18	3.3158

DISTRIBUTION BY COMMUNITY: Comm 0: 124 | Comm 1: 5 | Comm 2: 5 | Comm 5: 12

Temporal analysis: Phase 2 (Time and Rating)

PROBLEM:

High burstiness alone doesn't prove manipulation: a product could naturally receive clustered reviews after a media mention or seasonal trend. We need to detect anomalous rating shifts that deviate from historical patterns.

SOLUTION:

Baseline Deviation Analysis: We measure Change. If a product jumps from 1.5 to 4.0 stars something is happening.



Temporal analysis: Phase 2 (Time and Rating)

1



Group reviews into fixed time intervals

- for each product from Phase 1 Output

2



Calculate review count and rating distribution

- (mean, standard deviation, extremes) per window

3



Identify time intervals with anomalously high review activity

- compared to product's historical average

4



Flag bursts with extremely polarized ratings

- (nearly all 1-star or 5-star reviews)
- as potential coordinated attacks

5



Mean product rating computation before the burst

- and comparison with mean burst rating
- to see the difference between historical mean rating and suspicious coordinated ratings

Output



Visual timeline highlighting suspicious activity

- WHEN suspicious activity occurred
- and its rating pattern characteristics

Final Results

All shown cases are confirmed as:

- **Burst** (total_reviews >= ½ max_product_reviews)
- **Polarized** (mean_rating <=1.5 or >= 4.5)
- **Significantly Different** (rating_diff from baseline >=1)

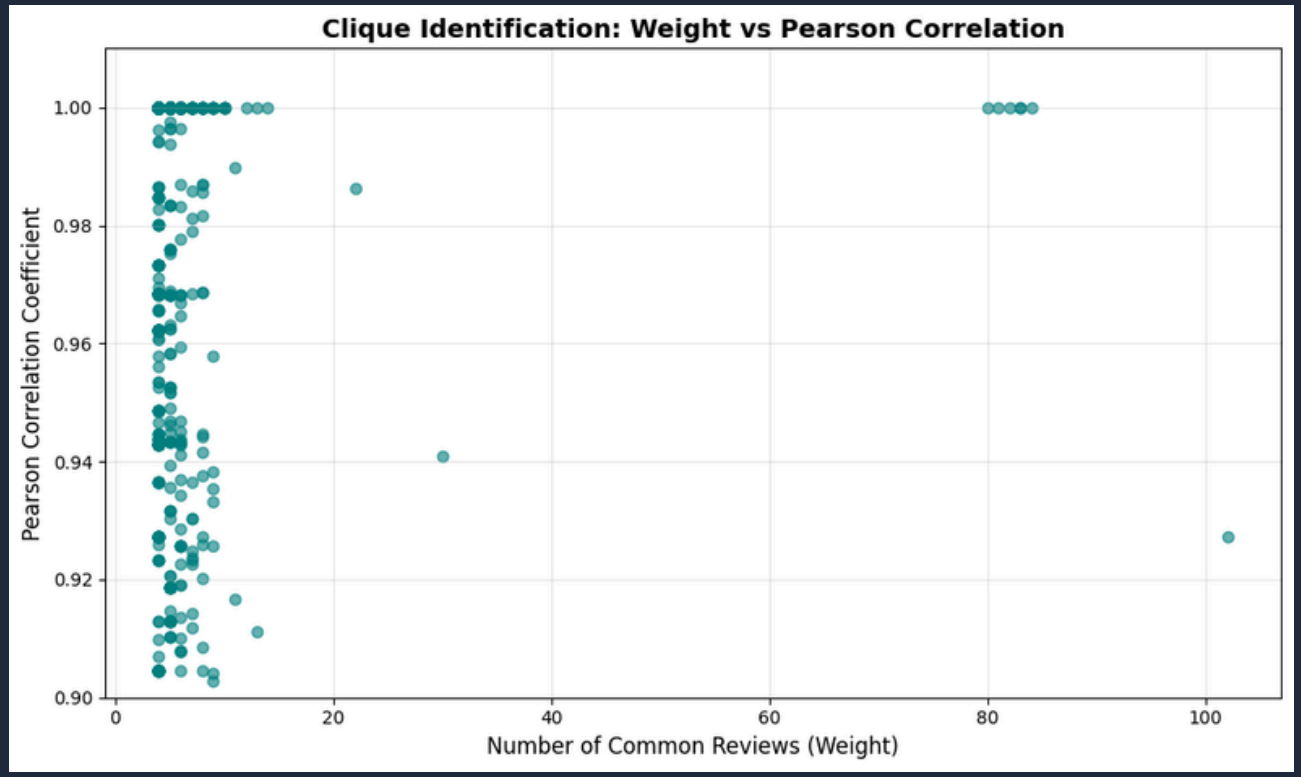
product_id	time	total_reviews	max_product reviews	mean_rating	baseline_prior_mean	rating_diff from baseline
1-DAY WINDOW						
53213	2006-04-26	9	9	4.7778	3.6981	+1.0797
10710	2006-01-16	4	5	5.0000	3.5217	+1.4783
10710	2006-01-11	3	5	5.0000	3.3333	+1.6667
...	...	...	...	...	...	...
2-DAY WINDOW						
10710	2006-01-15	5	6	5.0000	3.4545	+1.5455
10710	2006-02-18	5	6	5.0000	3.8971	+1.1029
92234	2006-03-17	5	5	4.8000	3.5556	+1.2444
266549	2006-04-23	5	5	4.8000	3.0000	+1.8000
13403	2006-01-02	4	6	5.0000	3.0000	+2.0000
...	...	...	...	...	...	...
179	2006-01-10	3	4	1.0000	5.0000	+4.0000
1-WEEK WINDOW						
13403	2006-01-02	10	12	4.8000	3.0000	+1.8000
53184	2006-01-01	10	12	4.7000	3.0000	+1.7000
10710	2006-01-15	9	15	5.0000	3.4545	+1.5455
266549	2006-04-22	7	7	4.7143	2.0000	+2.7143
40395	2006-01-09	5	5	4.8000	3.0000	+1.8000
...	...	...	...	...	...	...

Deep Dive – The 4-User Suspect Clique

- **Detection:** Starting from the Pearson-Weight scatter plot, we isolated 4 users connected by 6 edges with perfect correlation and 79 shared products.

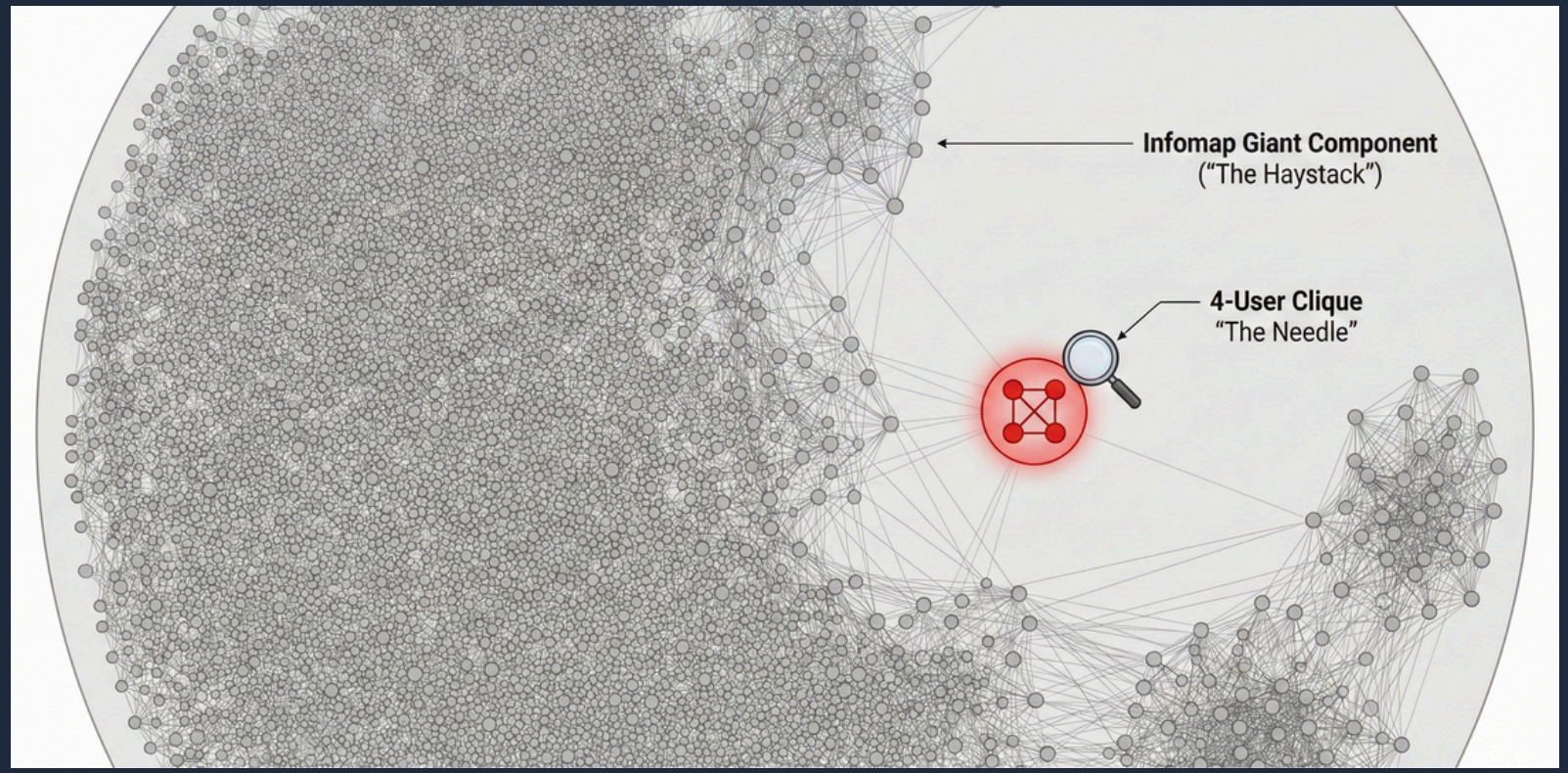
- **Validation:** Forensic analysis of these products identified 33 "bursty" items (synchronized timing) and 51 with significant rating deviation, confirming a coordinated manipulation attempt.

- **Intersection:** The "Smoking Gun" consists of 6 products flagged by both metrics, proving the core fraudulent activity that the global community analysis failed to isolate.



The Community Detection Paradox

- The 4-user clique was **absorbed into a massive Giant Component**, where global connectivity constraints merged this ultra-dense group with thousands of unrelated nodes.
- Once integrated into the macro-community, the exceptional local correlation was mathematically diluted, causing the internal Pearson average to drop and **masking the fraudulent signal**.



- **Conclusion:** This case proves that while global community detection maps the network, local edge analysis and behavioral metrics are essential to uncover high-fidelity anomalies that are topologically masked.

Thank you for your attention!

Any Questions?